# Human Performance & Limitations · Module D — Poisons & The Cabin Environment

**Chapter 10** — The physical environment around the pilot: humidity, temperature, vibration, glare and UV, and what happens when a pilot is incapacitated in flight — from a simple faint to a fit.

| BOOK | AUTHOR | EXAM | CHAPTER |
|---|---|---|---|
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*Plate 10.0 — The pilot's physical world: the heat, glare, dryness and vibration that quietly wear performance down.Open full-size image*

# § 23 Humidity

## 23.1 Why Humidity Matters Up High

**DGCA-QUOTED**

**Humidity may become an issue with high-altitude jet transport aircraft because of the low relative humidity at their operational altitudes.** The discomfort arising from low relative humidity may not imply physical indisposition.

**DGCA-QUOTED HUMIDITY RANGES**

- `40 – 60 %` **relative humidity is NORMAL.**
- **<** `20 %` may create **minor discomfort**, such as **skin, eye, nose, throat dryness**.

40–60% Normal relative humidity
< 20% Minor discomfort — skin, eye, nose, throat dryness
3–8% → 22% Modern humidifiers raise cabin RH from 3–8% to ~22%

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**PREVENTION & MANAGEMENT**

- **Overall dehydration can be prevented with adequate fluid intake.**
- **Diuretics like coffee or tea should be avoided.** They drive fluid out of the body via increased urination — the opposite of what is needed at altitude.

### 23.2 Aircraft Humidifier Systems

**DGCA-QUOTED**

The installation of humidifiers on aircraft **raises cabin/cockpit humidity**, but there are potential problems such as:

- **Weight penalty**,
- **Condensation**, and
- **Mineral contamination** that the designer must consider.

Aircraft such as the **Airbus A350, Boeing 787, and the future Boeing 777X** are compatible with onboard humidifiers, either in **crew areas or the entire cabin**. These technologies **increase humidity from 3–8 % to 22 %** and **reduce contaminants** in the air we breathe on a flight.

## § 24 Temperature

**DGCA-QUOTED**

**Temperature affects performance as follows:**

### 24.1 The Three Reference Temperatures — memorise verbatim

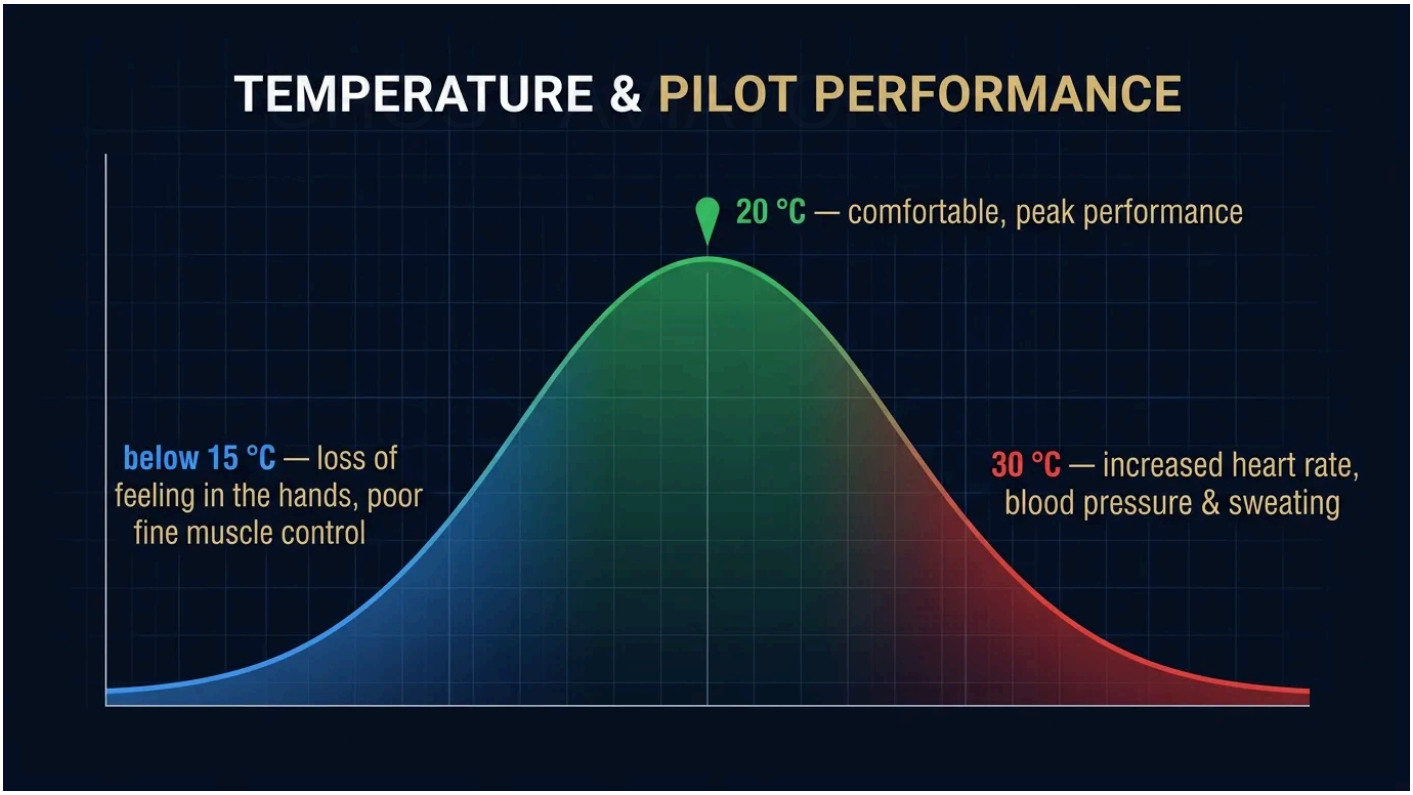
Temperature effects on pilot performance (DGCA verbatim)

| Temperature | Effect on Performance |
|---|---|
| 20 °C | **Comfortable temperature for most people in normal clothing.** |
| Above 30 °C | **Increased heart rate, blood pressure, and sweating.** |
| Below 15 °C | **Discomfort, loss of feeling in hands, poor control of fine muscle movement.** *(Exam trap: some printed question banks show this row as "> 15 °C". That is a misprint. 20 °C and 30 °C are both above 15 °C; the cold-hands effect starts BELOW 15 °C.)* |

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**COCKPIT-TEMPERATURE CONSIDERATIONS**

- **At 20 °C** the pilot is comfortable — fine motor skills, judgement and reaction time are optimal.
- **At 30 °C** the body's ANS responds to heat stress by raising HR, BP and triggering sweating to cool the skin. Cognitive performance starts to degrade.
- **In cold conditions** — far below the comfort baseline — the pilot loses sensation in the fingers, struggles with fine-motor tasks like switch selection or fast checklist scanning, and may shiver involuntarily.
- For both extremes, **cabin temperature management is a flight-safety task**, not merely a comfort one. Climate control should be set to keep cockpit temperature near 20 °C for sustained operations.




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*Figure 10.1 — Pilot performance peaks around 20 °C; both cold and heat degrade it in their own way.Open full-size image*

§ 25

# Vibration

## 25.1 Natural Resonance — Why Frequency Matters

**DGCA-QUOTED OPENING**

**Different parts of the body show a natural resonance at different periods of vibration.** For example:

- The **natural resonance of the eyeball is** `30 – 40 Hz`, and
- The **skull is** `1 – 4 Hz`.

**WHAT "NATURAL RESONANCE" MEANS IN PLAIN ENGLISH**

Every physical structure has a frequency at which it *wants* to vibrate — push it at that frequency and small inputs build up into huge oscillations (think of pushing a child on a swing in time with the swing's natural period). When external aircraft vibration matches a body part's resonant frequency, **that part suffers maximum disturbance**. The eyeball resonating at 30–40 Hz is precisely why **vision becomes blurred at certain helicopter or propeller-vibration frequencies**.

## 25.2 Effects of Vibration — Full DGCA Frequency Table

**MEMORISE THIS TABLE — DIRECT EXAM TARGET**

Effects of vibration include:

Vibration frequency vs body effect (DGCA verbatim)

| Frequency Range | Effect on the Body |
|---|---|
| `1 – 4 Hz` | **Interference with breathing; neck pain.** *(Skull resonance range.)* |
| `4 – 10 Hz` | **Chest and abdominal pain.** |
| `8 – 12 Hz` | **Backache.** |
| `10 – 12 Hz` | **Headache, eyestrain, throat pain, speech difficulty, muscle tension.** |
| `30 – 40 Hz` | **Interference with vision.** *(Eyeball resonance range.)* |

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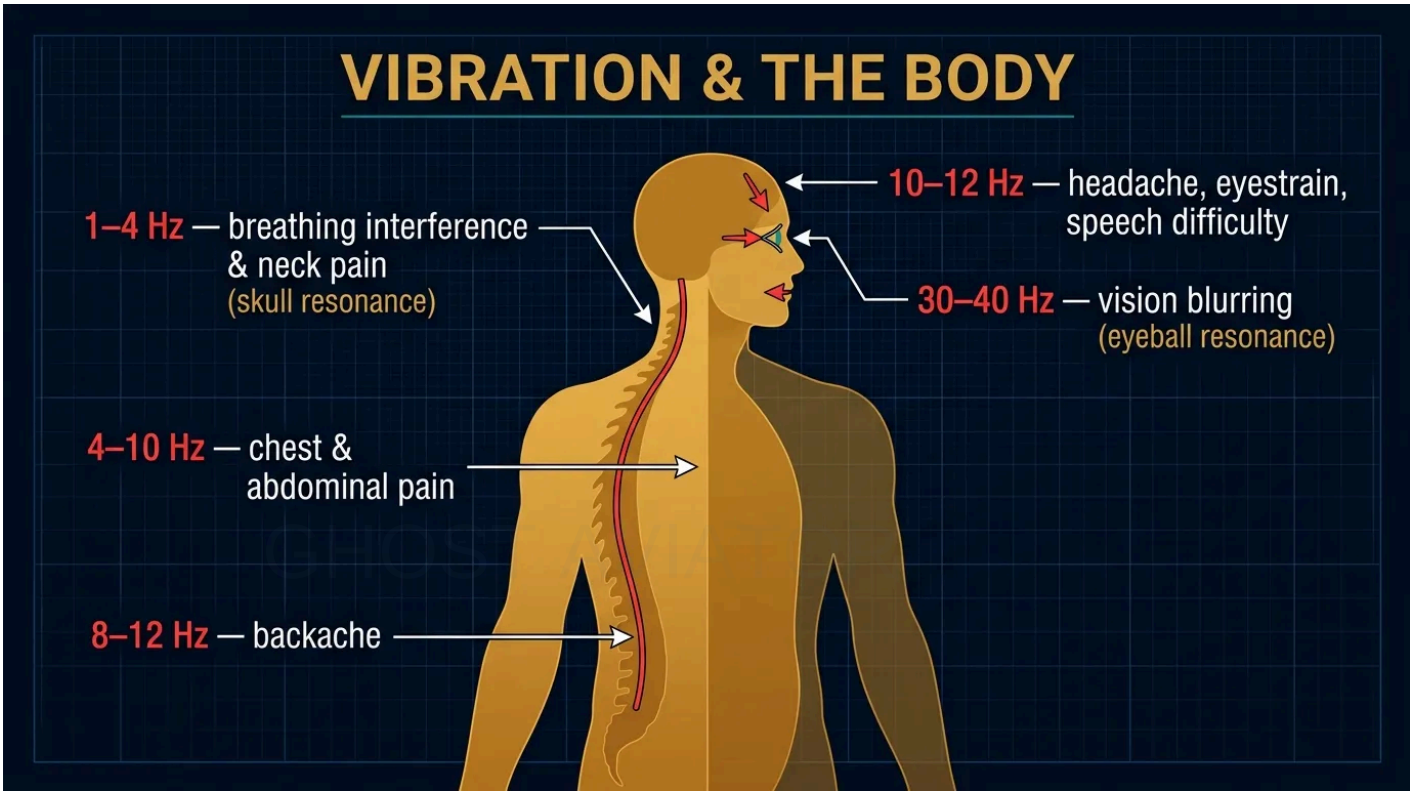



*Figure 10.2 — Each vibration frequency band affects a specific body region; the natural-resonance bands are the dangerous ones.Open full-size image*

**OPERATIONAL VIBRATION SOURCES IN AVIATION**

- **Helicopter rotors:** typically 4–7 Hz (main rotor) → chest/abdominal pain & backache range.
- **Piston / turboprop engines:** 10–30 Hz on the airframe → headache, eyestrain, vision interference.
- **Jet engine buffet:** higher-frequency, but resonant peaks in the seat may still strike critical body-part frequencies.
- **Turbulence:** low-frequency buffeting (1–10 Hz) → fits the skull / neck / chest range.

This is why aircraft seats incorporate **damping and isolation systems** — to keep the airframe's vibration spectrum from delivering energy at the worst body-resonance frequencies.

## § 26 Glare & UV Radiation

**DGCA-QUOTED**

**UV radiation from sunlight can cause visual fatigue, as well as affect visual health.**

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**WHY GLARE AND UV DESERVE THEIR OWN SECTION**

At cruise altitudes (FL300+), the protective filtering of the atmosphere is reduced. Cockpit windows admit **significantly more UV-A and UV-B** than ground-level glass would. Over a long career, this contributes to:

- **Visual fatigue** — the immediate effect of glare on the working pilot. Squinting, eyestrain, reduced contrast sensitivity, headache.
- **Cumulative visual health damage** — cataract formation, macular degeneration risk, conjunctival/skin damage.

**PRACTICAL PILOT PRECAUTIONS**

- Use **aviation-grade polarised or non-polarised UV-blocking sunglasses** (note: certain LCD instruments may distort under polarised lenses — check before adoption).
- Use **cockpit sun-visors** appropriately — don't leave them stowed when the sun is over the nose.
- For long-haul crews, schedule **vision rest breaks** on autopilot legs.
- Periodic **aero-medical eye examinations** — already mandated by DGCA — pick up early cataract / retinal changes.

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# § 27 INCAPACITATION IN FLIGHT

## 27.1 Why Pilot Incapacitation is a Flight-Safety Issue

**DGCA-QUOTED INTRODUCTION — VERBATIM**

**The risk of seizure in flight is obvious.** Incapacitation is in most cases:

- **Sudden**,
- **Unpredictable**,
- **Unavoidable**,
- **Prolonged**,
- **Complete**, and
- **Potentially more frequent in the stressful flying environment**,

... and constitutes a **direct threat to the health and safety**.

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**DGCA-QUOTED PREVENTION MEASURE**

**Periodical medical examinations minimize the risk of incapacitation in flight. The frequency of medical checks increases with advancing age.**

| | |
|---|---|
| CLASS-1 MEDICAL (CPL / ATPL), UP TO AGE 60 | **12 months** |
| CLASS-1 MEDICAL, ABOVE 60 | **6 months** |
| CLASS-1, SINGLE-CREW OPERATIONS, ABOVE 40 | **6 months** |
| CLASS-2 MEDICAL (PPL / SPL), UP TO AGE 50 | **2 years** |
| CLASS-2 MEDICAL, ABOVE 50 | **1 year** |

*(DGCA CAR Section 7 Series C Part I, para 5.1, effective 15 Nov 2025.)*

## 27.2 Obvious Incapacitation

**DGCA-QUOTED DEFINITION**

**Obvious incapacitation** normally refers to a state in which **all of a crew member's physical or mental functions are lost**, including:

- **Loss of consciousness**, OR
- **Being unable to move while retaining consciousness**,

... **rendering them completely unable to carry out their duties**.

**ONSET PATTERNS & PRESENTATION**

- Obvious incapacitation can sometimes involve **convulsions**,
- OR the victim may **lapse into unconsciousness several minutes after the incapacitation occurs**.

**DGCA-QUOTED CAUSES OF OBVIOUS INCAPACITATION**

- **Cardiac arrest**
- **Myocardial infection** *(textual variant — "myocardial infarction" — heart attack)*
- **Interracial hemorrhage** *(intended: "intracranial haemorrhage" — bleed in the brain)*
- **Cerebral apoplexy** *(stroke)*
- **Epilepsy**

*The source PDF uses the phrasings shown — students should be aware of the modern medical terminology in parentheses.*

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## 27.3 Subtle / Insidious Incapacitation

**DGCA-QUOTED DEFINITION**

**SUBTLE** (develops slowly and gradually), incapacitation refers to a state of a **partial or temporary loss of physical or mental function** which manifests itself in the form of:

- **Partial paralysis**,
- **A dulling of perception, judgment, or responses** or lack thereof,
- **A state of absence of mind, distraction of attention**,
- **Trouble with speech**,
- **Inadequate responses**,
- **Meaningless utterances**, etc.

**THE HIDDEN-DANGER WARNING — DGCA VERBATIM**

**Particular attention must be paid to the fact that a crew member may become incapacitated EVEN THOUGH THEIR APPEARANCE IS NO DIFFERENT FROM NORMAL.**

**POSSIBLE CAUSES OF SUBTLE INCAPACITATION — DGCA LIST**

- **Temporary hypoglycemia** (low blood sugar — pilot skipped meals)
- **Reduction in blood pressure**
- **Cerebropathy or psychopathy**
- **Excessive muscular fatigue**
- **Excessive drinking** (see §14)
- **Insufficient sleep**
- **Emotional instability**
- **Toothache**
- **Stomachache**
- **Headache**

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## 27.4 Obvious vs Subtle — Why Subtle is MORE Dangerous

**DGCA-QUOTED — THE COUNTERINTUITIVE TRUTH**

**Since other crew members are often unable to detect subtle incapacitation quickly, from a flight safety point of view, it is possible that SUBTLE INCAPACITATION MAY LEAD TO A SITUATION OF COMPARATIVELY GREATER DANGER THAN OBVIOUS INCAPACITATION.**

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**WHY THIS MATTERS IN TWO-CREW OPERATIONS**

With **obvious incapacitation**, the other crew member sees the captain slumped or convulsing, declares emergency, takes control, gets the aircraft on the ground. The hand-off is instant.

With **subtle incapacitation**, the captain is still moving switches, talking on the radio, even responding to questions — but his judgement is impaired. He may set the wrong altitude, mis-tune a NAV, mishear an ATC clearance, or fail to initiate a missed approach. The other pilot may not detect this until *after* the wrong outcome unfolds. This is the rationale behind **two-crew CRM**: a healthy independent cross-check that catches subtle drift before it becomes an accident.

### *OBVIOUS Incapacitation*

- **All functions lost** — LoC or paralysed-conscious
- May include convulsions / delayed LoC
- **Detected instantly** by the other crew
- Causes: cardiac arrest, MI, intracranial bleed, stroke, epilepsy
- **Handover happens fast**

### *SUBTLE Incapacitation*

- **Partial or temporary** loss of function
- Appearance may be normal
- **Often undetected for some time** by the other crew
- Causes: hypoglycemia, ↓ BP, fatigue, sleep loss, drinking, emotional/dental/stomach/headache
- **Greater overall flight-safety danger**

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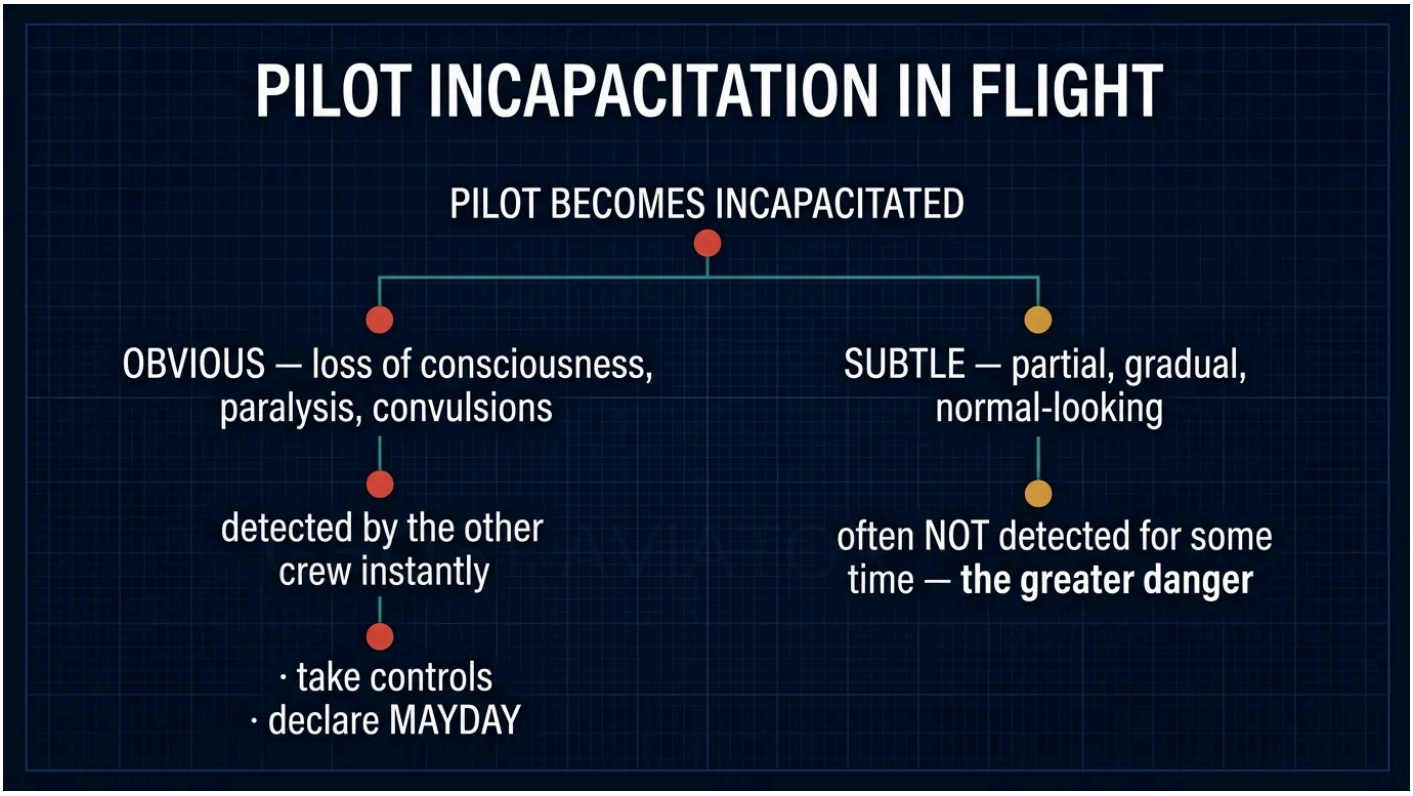



*Figure 10.3 — Pilot incapacitation: recognising it and the other crew member's response.Open full-size image*

## § 28 FITS & FAINTS

### 28.1 Epilepsy — Grand Mal & Petit Mal

**DGCA-QUOTED DEFINITION**

A **fit or seizure is usually referred to as "epilepsy"**. A fit or a seizure is **not a specific disease** but **a set of signs or symptoms in response to a disturbance of the electrical activity in the brain**.

***Grand Mal Epilepsy***

- Manifests as a **generalized seizure**
- **Associated with a transient loss of consciousness**
- May be associated with a **prodromal phase** *(warning aura before the seizure)*
- Normally accompanied by **convulsions and uncontrolled physical movement**

***Petit Mal Epilepsy***

- Also a **generalized seizure**
- **Not associated with a loss of consciousness**
- **Petit Mals are a MINOR attack**

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- Often presents as brief "absence spells" — staring, blanking out for a few seconds

**DGCA-QUOTED — THE ABSOLUTE BAR**

**A seizure may or may not be associated with a loss of consciousness**... but **ANY FIT, MAJOR OR MINOR, IS ASSOCIATED WITH AN UNPREDICTABLE LOSS OF CONSCIOUSNESS AND IS THEREFORE AN ABSOLUTE BAR TO THE HOLDING OF A FLYING LICENCE.**

**PILOT-MEDICAL IMPLICATION**

This is one of the very few *absolute* medical disqualifications in aviation. Even controlled epilepsy on medication — even a single documented seizure event in adult life — is grounds for medical refusal. The reason is the word "**unpredictable**" — DGCA cannot certify the probability of an in-flight seizure as acceptably low.

## 28.2 Faint & Vasovagal Syncope

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**DGCA-QUOTED DEFINITION**

**Faint is a common cause of a loss of consciousness in adults.** The most common causes of faints are:

| Common causes of faint — DGCA verbatim list | |
|---|---|
| **#** | **Cause** |
| 1 | **Standing up quickly after prolonged sitting especially when hot or dehydrated** |
| 2 | **A sudden shock** |
| 3 | **Loss of blood after an accident** |
| 4 | **Lack of food or fluid** |
| 5 | **Other physiological stress** |

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**DGCA-QUOTED — SYNCOPE & VASOVAGAL SYNCOPE**

**Syncope is a temporary but sudden loss of consciousness when blood flow to the brain is compromised.** In young individuals, **fear, anxiety, sight of blood, etc., can result in a temporary loss of consciousness**. This is referred to as **VASOVAGAL SYNCOPE**.

Frequently, syncope is associated with symptoms like:

- **Light-headedness**,
- **Muscle weakness**, and
- **Dizziness** before the actual fainting occurs.

**DGCA-QUOTED — FLIGHT-LICENCE IMPLICATION**

**A faint has NO SIGNIFICANCE as far as future flying is concerned, so long as the cause is clearly understood.**

**EPILEPSY VS FAINT — THE CRITICAL COMPARISON**

This is the most frequently tested contrast in DGCA HPL papers. Memorise it:

Epilepsy vs Faint (Syncope) — side-by-side

| Parameter | Epilepsy (Grand/Petit Mal) | Faint (Syncope) |
|---|---|---|
| Underlying cause | **Disturbance of brain electrical activity** | **Compromised blood flow to the brain** |
| Loss of consciousness | Often (Grand Mal) / Absent (Petit Mal) | **Yes, temporary & sudden** |
| Convulsions / uncontrolled movement | **Yes (Grand Mal)** | No |
| Predictability | **Unpredictable** | Cause is usually identifiable (heat, hunger, fear, blood loss) |
| Typical warning signs | Prodromal aura (Grand Mal) | **Light-headedness, muscle weakness, dizziness** |
| Effect on flying licence | **ABSOLUTE BAR — no licence** | **No significance — fit to fly if cause is clear** |

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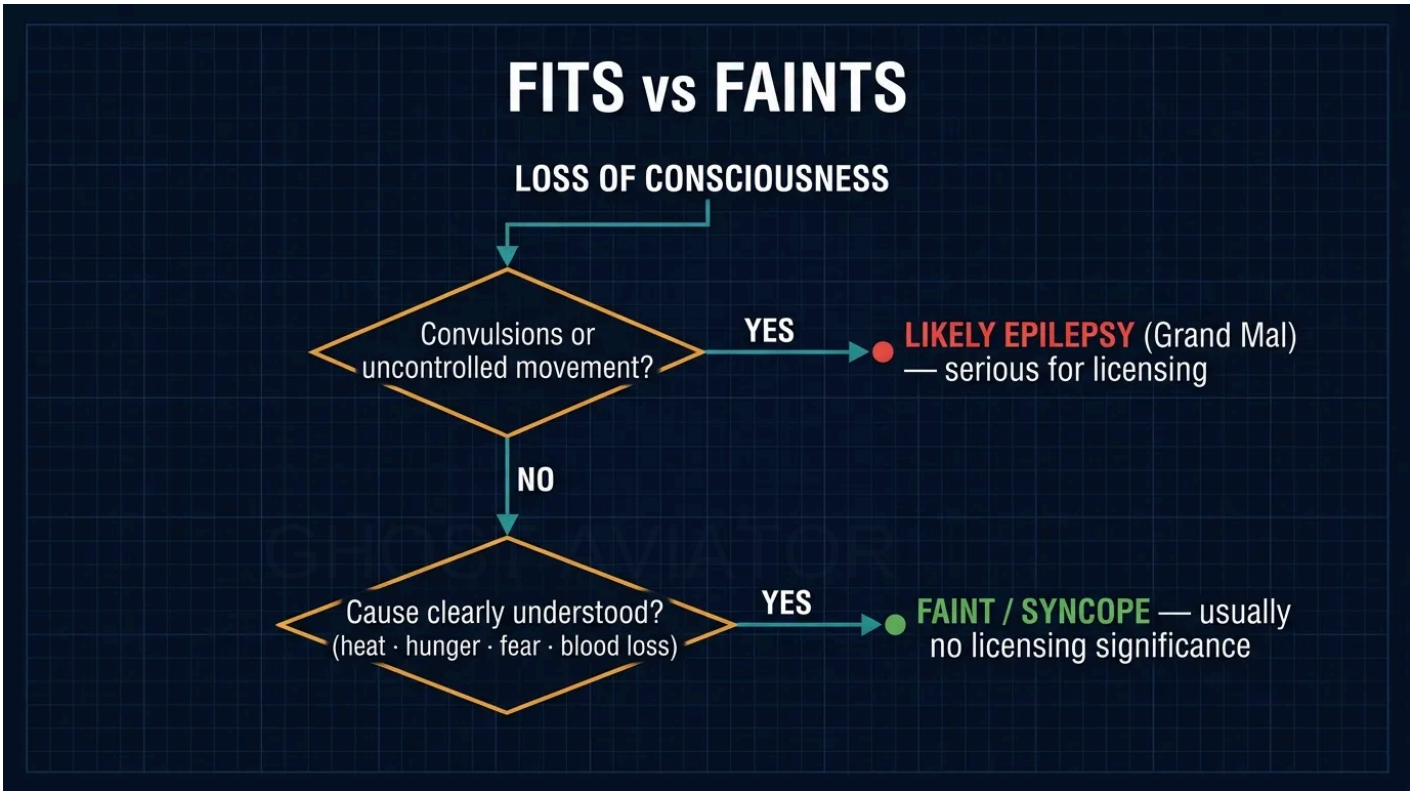



*Figure 10.4 — Fits and faints: causes of syncope and how epilepsy differs from a simple faint.Open full-size image*

✦ END OF CHAPTER 10 ✦

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